

September 2026

Evaluating LLM-Based Approaches for Gene Normalization on the NLM-Gene Benchmark

Presented by : **Xiaomeng Li**
Supervised by : **Nona Naderi**
Louise Deléger
Arnaud Ferré



Gene Normalization: Why Context Matters

Gene normalization links gene or protein mentions to **NCBI Gene** entries

Recently, it has been reported that using multiple signals, murine and human B cells secrete several cytokines with pro-inflammatory and immunoregulatory properties. We present the first comprehensive analysis of 24 cytokines, chemokines, and hematopoietic growth factors production by purified **human** peripheral blood B cells (CD19+), and naive (CD19+CD27-) and memory (CD19+CD27+) B cells in response to direct and exclusive signaling provided by toll-like receptor (TLR) ligands Pam3CSK (TLR1/TLR2), Imiquimod (TLR7), and GpG-ODN2006 (TLR9). All three TLR ligands stimulated B cells (CD19+) to produce cytokines IL-1 α , IL-1 β , IL-6, TNF- α , IL-13, and IL-10, and chemokines MIP-1 α , MIP-1 β , **MCP-1** IP-10, and IL-8. However, GM-CSF and G-CSF production was predominantly induced by TLR2 agonist. Most cytokines/chemokines/hematopoietic growth factors

Abstract example

NLM-Gene corpus · PMID: 20821041

Context example

Normalized entity

MCP-1 in **human** macrophages

CCL2 · GeneID **6347**

MCP-1 in **mouse** macrophages

Ccl2 · GeneID **20296**

Which LLM-based approach works best for gene normalization?

Four Ways to Use an LLM

01.

**Direct GeneID
generation**

Prompt → GeneID

02.

**Gene name
generation**

*Prompt → Gene name →
NCBI mapping*

03.

**Dictionary-based
selection**

*Lexical retrieval →
Candidates → Qwen*

04.

**SapBERT-based
selection**

*Embedding retrieval →
Candidates → Qwen*

You are doing biomedical gene normalization.

Task: Given a gene/protein mention, its local context, and a list of candidate NCBI Gene IDs, choose the most likely correct candidate.

Important rules:

- Choose exactly one option from the candidate list.
- Do not invent a new Gene ID.
- Only output the option letter.

Mention: MCP-1

Context:

... increased expression of MCP-1 was observed in inflammatory cells ...

Candidates:

A. GeneID: 6347 | Species: human | Matched term: MCP-1 / CCL2

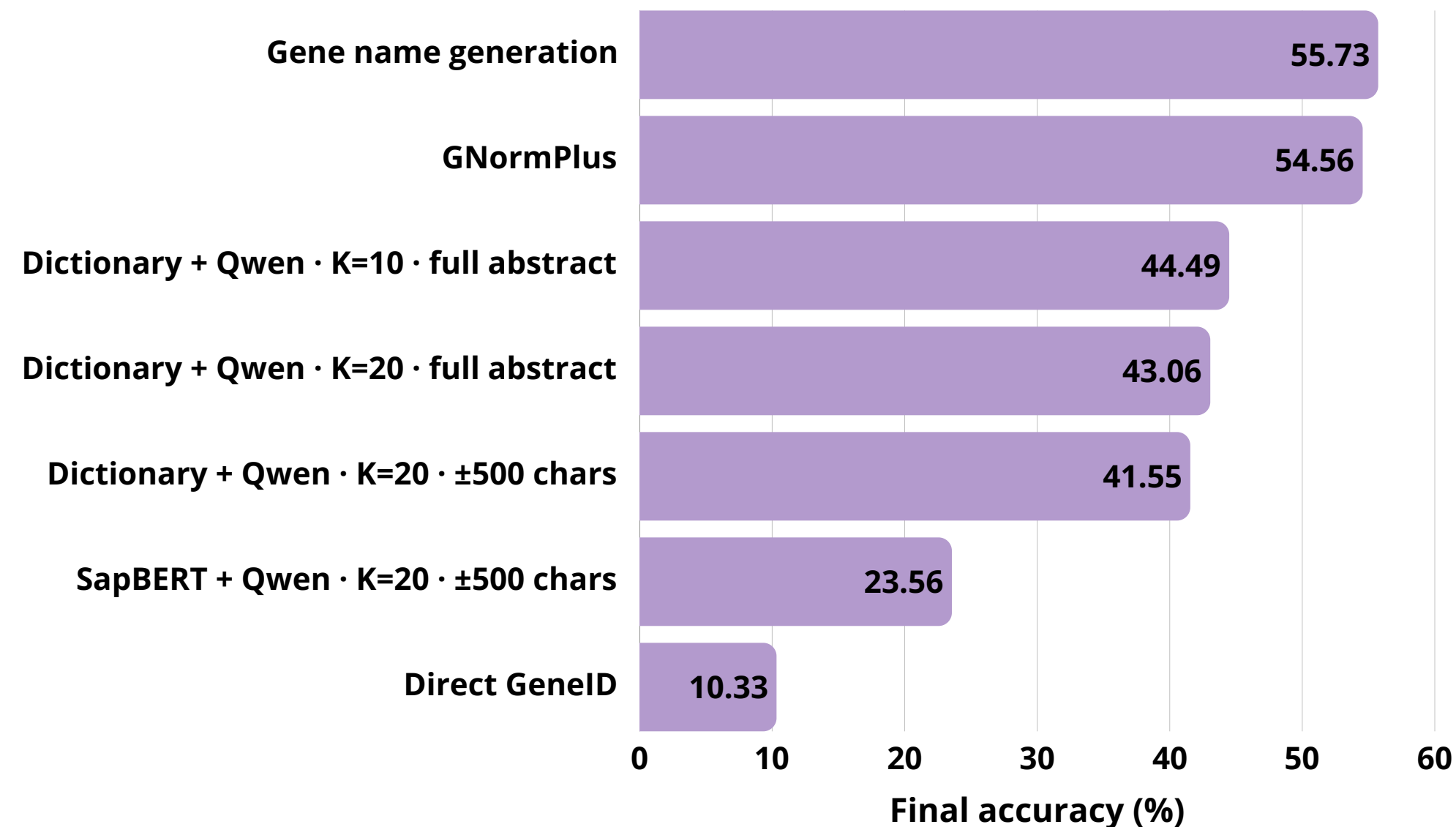
B. GeneID: 20296 | Species: mouse | Matched term: MCP-1 / Ccl2

C. GeneID: 24770 | Species: rat | Matched term: MCP-1 / Ccl2

Answer:

Example prompt for approach 3

Gene Name Generation Gives the Best LLM-Based Result



- **55.73%:** Gene name generation
- **10.33%:** Direct GeneID generation
- **Retrieval remains the main bottleneck**

Generating a gene name before database mapping is more reliable than directly predicting a GeneID.

Future Work:
improve candidate retrieval and species disambiguation.

Final accuracy: Percentage of all test mentions assigned the correct NCBI GeneID

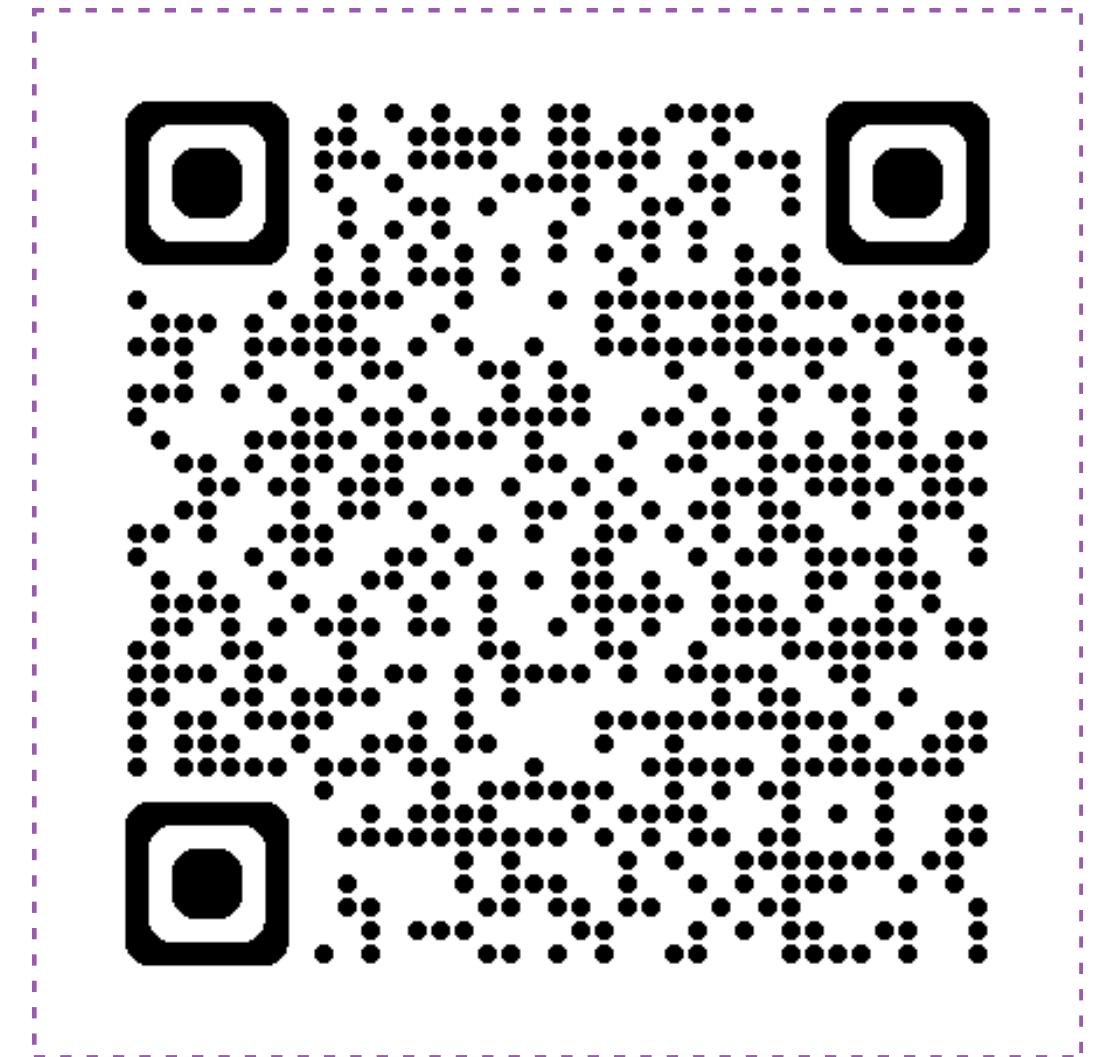
K: Number of retrieved candidate genes provided to Qwen for each mention

September 2026

Thank You

Presented by :
Xiaomeng Li

Supervised by :
Nona Naderi
Louise Deléger
Arnaud Ferré



Codes & Paper